

dea-tools: geospatial analysis of satellite data using Digital Earth Australia, Open Data Cube, and Xarray.

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Summary

dea-tools is an open-source Python package for geospatial analysis of satellite data using Digital Earth Australia, [Open Data Cube](#) (ODC) and xarray ([Hoyer & Joseph, 2017](#)). It provides a broad set of utilities for loading, visualising, transforming, and analysing Earth Observation (EO) data across space and time

The package includes tools for:

- **Data handling:** Load and combine DEA data products, manage projections and resolutions.
- **Visualisation:** Create static and interactive maps, RGB plots, and animations.
- **Remote sensing indices:** Calculate band indices such as NDVI, NDWI, and more.
- **Spatial and temporal analysis:** Apply raster/vector operations, extract contours, compute temporal stats, and model change over time.
- **Machine learning and segmentation:** Train and apply classifiers, or run image segmentation workflows.
- **Parallel processing:** Set up Dask clusters for scalable processing of large datasets.
- **Domain-specific tools:** Analyse coastal change, intertidal zones, land cover, wetland and waterbody dynamics, and climate datasets.

Statement of need

Satellite remote sensing offers an unparalleled resource for

Features

Analytical tools native to Xarray

Flexible and powerful functionality for scientific analysis with xarray objects:

- Extracting time series statistics, using the module `dea-tools.temporal`:
 - Generic summary statistics on any time series with `temporal_statistics`
 - Phenology with `xr_phenology`
 - Linear regression with `xr_regression`
- Validation tools within `dea-tools.validation`:
 - Compute common statistical metrics with `eval_metric`
 - Generate random points over an xarray object with `xr_random_sampling`

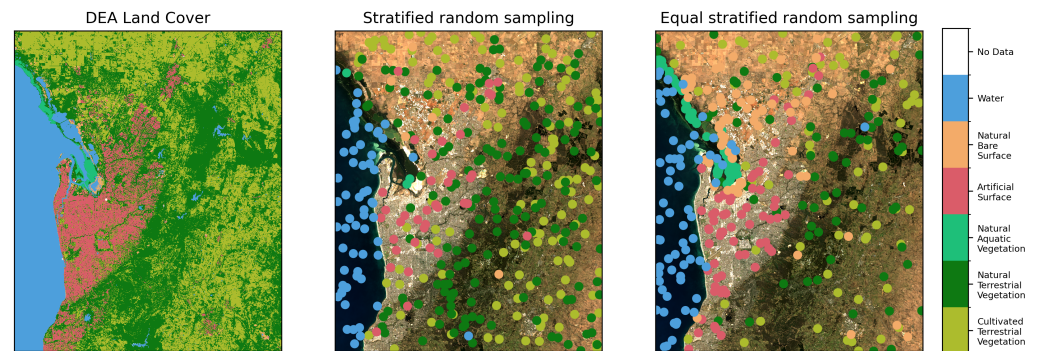


Figure 1: Example extraction of random samples from a classified xarray.DataArray using two of the sampling strategies available in the xr_random_sampling function.

- Spatial analysis tools within dea-tools.spatial:
 - Vectorise xarray.DataArrays into geopandas.GeoDataFrames with xr_vectorise
 - Rasterize geopandas.GeoDataFrames into xarray.DataArrays with xr_rasterize
 - Extract subpixel contours from xarray.DataArrays with subpixel_contours
 - Interpolate point data stored in a geopandas.GeoDataFrame into an xarray.DataArray with xr_interpolate
- Xarray wrappers for the full machine learning pipeline in dea-tools.classification:
 - prepare data for ML predictions with sklearn_flatten and sklearn_unflatten
 - Fit models with fit_xr
 - ML predictions with predict_xr
 - Handle spatial autocorrelation in cross-validation workflows with spatial_train_test_split and SKCV (spatial k-fold cross validation)

Plotting

The dea-tools.plotting library has extensive functionality for generating beautiful imagery from xarray objects. - RGB plots for true and false colour composite images with rgb - Highly customisable animations with xr_animation

Integration with Open Data Cube libraries

dea-tools works naively with odc-geo

Acknowledgements

Thanks

References

- Hoyer, S., & Joseph, H. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, 5(1). <https://doi.org/10.5334/jors.148>